NATURAL LANGUAGE PROCESSING

Skill Signal

SkillSignal: An Explainable Resume–Job Matching System with a Writing-Style Fairness Audit

Programme	BBA (Artificial Intelligence)
Academic Year	2026-27
Team Members	**1. Ayush Sahoo - 25030422031** **2. Yana Khandelwal - 25030422141** **3. Anushka Thakur - 25030422019**
Faculty	**Dr. Nabanita Choudhury**, Assistant Professor

SYMBIOSIS ARTIFICIAL INTELLIGENCE INSTITUTE (SAII)

Constituent of Symbiosis International (Deemed University)
Lavale, Pune - 412115

1. Abstract

Resume screening tools often rank candidates by how closely their words match a job description. That is unfair to people who have the same skills but describe them differently. SkillSignal is a small resume–job matching system we built to see how much writing style alone can move a match score. It has two models: Model A uses TF-IDF with cosine similarity, and Model B uses sentence embeddings (all-MiniLM-L6-v2) with cosine similarity. We tested both on a public dataset (Resume–Job Description Fit, Hugging Face), using Business Analyst postings and the metrics Precision@5, Recall@5 and NDCG@5. For the fairness audit we wrote 15 fictional candidate profiles, each in a concise and a verbose version that list exactly the same skills. On the real data the two models tied on Precision@5 (0.467) and Recall@5 (0.406), while the embedding model was slightly ahead on NDCG@5 (0.700 against 0.677). That comparison rests on only 3 job postings, so it should be read lightly. In the audit, rewriting the profiles shifted TF-IDF scores by 0.187 on average and embedding scores by 0.080. Rankings changed for 93.3% of profiles with TF-IDF and 80.0% with embeddings, and our dictionary-based skill extractor missed every skill in all 15 verbose profiles. Both models turned out to be sensitive to wording. Embeddings were steadier, but they were not immune.

2. Introduction

Background and context. Companies get a lot of resumes for every opening. To save time, many of them use software that compares each resume with the job description and returns a match score. The simpler versions of this software mostly count shared words, so a resume that copies the language of the job post tends to score higher.

Motivation. Two people can have the same skills and still write about them differently. One might put “SQL, Excel, requirements gathering”, while another writes “built reports from large datasets and captured business needs from several teams”. If the software only rewards matching words, the second person can end up lower in the list for no good reason. We wanted to measure how big this effect really is rather than just assume it.

Role of NLP. NLP shows up in three places in this project. First, text cleaning and TF-IDF turn text into numbers based on how important each word is. Second, sentence embeddings try to capture meaning beyond the exact words. Third, a skill dictionary with phrase matching shows which skills were found and which were missing. Running these methods on the same text lets us see which one leans harder on exact wording.

3. Problem Statement and Objectives

Problem statement. Resume–job matching systems may favour candidates whose wording is close to the job description, and may overlook equally qualified candidates who express the same skills in a different way.

Research question. Does the way a candidate describes their experience change their resume–job match score when their underlying skills stay the same?

Objectives.

- Build two ways of matching: TF-IDF with cosine similarity (Model A) and sentence embeddings with cosine similarity (Model B).
- Pull out skills using a fixed skills dictionary, and show the matched and missing skills for each candidate.
- Test both models on real, labelled resume–job pairs using Precision@K, Recall@K and NDCG@K.
- Make a controlled set of paired concise and verbose profiles with known skills, and measure how much the score and rank change.
- Look at the error cases, explain the likely reasons in the text, and be clear about the limitations.

4. Dataset Description

We used two datasets. Dataset A is real text, and we used it to test the pipeline and measure matching quality. Dataset B is synthetic and was used only for the writing-style audit, because that is the one place where we can be sure two resumes have equal qualifications.

4.1 Dataset A: Real resume–job data

- **Name and source:** "resume-job-description-fit" (user cnamuangtoun) on Hugging Face Datasets, loaded with load_dataset("cnamuangtoun/resume-job-description-fit"). The usage terms still need to be checked on the dataset page.
- **Size:** 6,241 training rows and 1,759 test rows, so 8,000 resume–job pairs in total. We used only the test split.
- **Fields:** resume_text, job_description_text and label (No Fit, Potential Fit, Good Fit).
- **Language:** English.
- **Class distribution:** No Fit is the biggest class, at about half of all rows. Table 1 has the counts.

Table 1: *Label distribution of Dataset A*

Split / subset	Rows	No Fit	Potential Fit	Good Fit
Train	6,241	3,143	1,556	1,542
Test	1,759	857	444	458
Business Analyst subset (from test)	240	120	59	61

Subset used. To stay within one job family (as we planned), we picked the test rows whose job description contains "business analyst". That gave 240 rows and 10 unique job descriptions. For ranking we kept only jobs with at least 10 candidates, which left 7 jobs. Jobs with no Good Fit candidate had to be skipped because Recall cannot be measured for them, so 3 jobs remained for the final metrics. For the metrics, "Good Fit" counted as relevant (1), and "Potential Fit" and "No Fit" counted as not relevant (0). For NDCG we used graded relevance instead: Good Fit = 2, Potential Fit = 1, No Fit = 0.

4.2 Dataset B: Controlled synthetic writing-style experiment

Dataset B was made by us and is entirely synthetic. It contains fictional profiles only, so its results describe this controlled experiment and not real applicants.

Table 2: *Summary of the synthetic Dataset B*

Item	Detail
Profiles	15 fictional profiles (Profile_1 to Profile_15)
Versions per profile	2: a concise version and a verbose version
Target job	One written Business Analyst job description (job_description_B)
Skills list	8 required skills: requirements gathering, stakeholder management, process improvement, agile, Excel, documentation, communication, data analysis
Skills per profile	3 to 8; Profile_1, Profile_7 and Profile_15 have all 8
Ground truth	The true skill list of each profile is stored, so both versions are known to be equivalent

5. Data Preprocessing

We kept the preprocessing simple on purpose, so it is easy to explain and repeat.

- **Lowercasing and punctuation handling:** every text is converted to lowercase, and every character that is not a letter, digit or space is replaced with a space using a regex. This also gets rid of URL symbols, bullet signs and other special characters. Extra spaces are then collapsed into one.
- **Stop-word removal:** English stop words are removed inside TfidfVectorizer (stop_words="english") for Model A only. We did not use stemming or lemmatization.
- **Missing or noisy text:** every value is converted with str() so that missing values do not crash the code. For the real data we used only the first 3,000 characters of each resume and job description, which keeps the run time low and stays within the input length of the embedding model.
- **Raw text for embeddings:** Model B gets the original, uncleaned text, because sentence encoders make use of word order and punctuation.
- **Not handled:** we did not treat emojis, hashtags or mentions separately. The data is English only, so multilingual or code-mixed text was not a concern.

6. Feature Representation / Embeddings

TF-IDF (Model A). Each text becomes a vector in which a word gets a high weight if it is frequent in that text but rare in the other texts. We picked TF-IDF as the baseline because it is fast, simple and easy to inspect, since we can list the overlapping words that produced a score. On the real data, the vectorizer was fitted separately for each job together with its candidate resumes (maximum 3,000 features). For the audit, it was fitted on the job description plus the 15 profiles of one version.

Sentence embeddings (Model B). We used the pretrained model all-MiniLM-L6-v2 from the sentence-transformers library. It turns a whole text into one dense vector, so texts with similar meaning can end up with similar vectors even when the words differ. That is why we expected it to cope with rewording better than TF-IDF. We did no training or fine-tuning.

Similarity. For both models the match score is the cosine similarity between the job vector and the resume vector. It is a text-similarity score and not a probability of being hired.

7. Proposed Methodology / System Architecture

Figure 1 shows the workflow from start to finish. A job description and resume texts go in, and Model A, Model B and the skill extractor each process them. Their outputs are put together in a match report (scores, ranking, matched and missing skills), which then feeds the evaluation on real data and the writing-style audit on synthetic data.

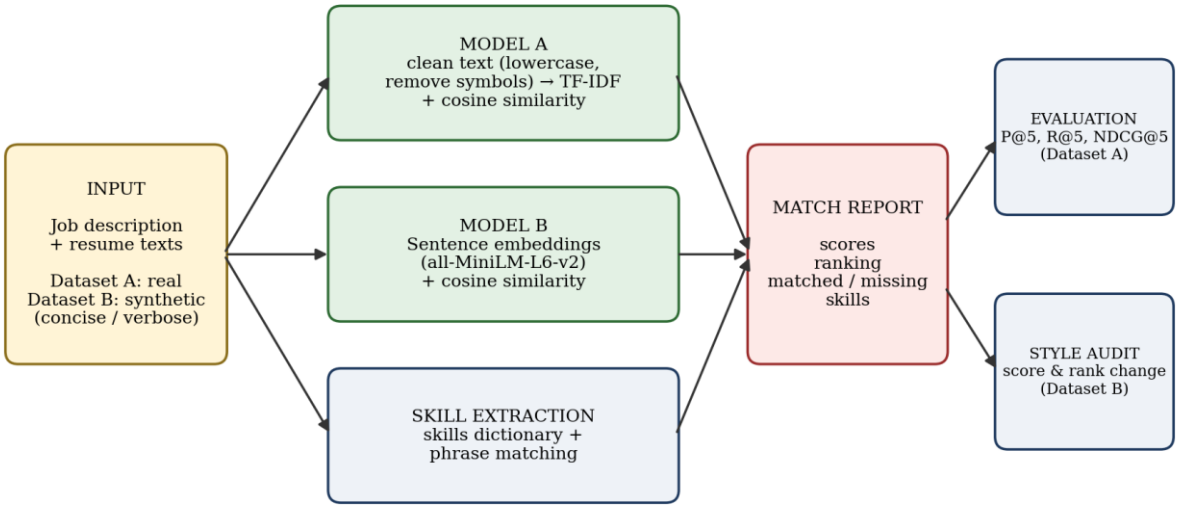


Figure 1: *End-to-end workflow of SkillSignal*

Steps followed in the project:

- Load Dataset A and select the Business Analyst job postings.
- Fix the required-skills dictionary before looking at any results.
- Clean the text, then compute TF-IDF scores (Model A) and embedding scores (Model B).
- Rank the candidates for each job by score and calculate Precision@5, Recall@5 and NDCG@5 against the dataset labels.
- Create Dataset B: 15 profiles, each with a concise and a verbose version using the same skills.
- Score both versions with both models, then measure the score change and rank change.
- Check which true skills the skill extractor still finds in the verbose versions, and study the errors.

8. NLP Model / Technique Used

Model A: TF-IDF + cosine similarity. This is our lexical baseline. It rewards exact word overlap between the resume and the job description.

Model B: Sentence embeddings + cosine similarity. A semantic model (all-MiniLM-L6-v2). It can give credit for similar meaning, but it is a black box. We can only explain it through matched skill phrases and controlled paraphrase tests, not through what happens inside the model.

Skill extraction. A skills dictionary maps each skill to a list of words or phrases (for example, "agile" is matched by "agile" or "scrum"). The cleaned text is searched for these phrases with simple substring matching. The same function runs on the job description (to get the required skills) and on each resume (to get the candidate skills). Matched skills are required ∩ candidate skills, and missing skills are required − candidate skills. The full dictionary is in Annexure A.

Training. No model was trained. TF-IDF is fitted on the texts of each job group, and the embedding model is used as it comes.

Table 3: *Main settings used in the experiments*

Setting	Value
Embedding model	all-MiniLM-L6-v2 (sentence-transformers)
TF-IDF	TfidfVectorizer, English stop words; max_features = 3,000 (real data)
Similarity	Cosine similarity
Text length (real data)	First 3,000 characters of each text
Minimum candidates per job	10
Ranking cut-off (K)	5
Relevance labels	Binary: Good Fit = 1, others = 0. Graded (NDCG): 2 / 1 / 0

Evaluation measures. Precision@5 is the fraction of the top 5 candidates that are Good Fit. Recall@5 is the fraction of all Good Fit candidates for that job that show up in the top 5. NDCG@5 uses the graded labels to check how well the ranking puts better candidates near the top. For the audit we used the mean absolute score change between the concise and verbose version of each profile, and the percentage of profiles whose rank (among the 15 profiles) changed after rewriting.

9. Implementation

- **Language and libraries:** Python in a Google Colab notebook, using datasets, sentence-transformers, scikit-learn (TfidfVectorizer, cosine_similarity), pandas, numpy, re and matplotlib.
- **Major components:** (1) data loading and label mapping, (2) Business Analyst subset selection, (3) text cleaning, (4) skill extraction with the skills dictionary, (5) TF-IDF and embedding scoring, (6) ranking metrics, (7) a generator for the concise and verbose

synthetic profiles, (8) the writing-style audit and plots, and (9) a printed candidate report with matched and missing skills.verbose versions, (8) writing-style audit and plots, (9) a printed candidate report with matched and missing skills.

- **Synthetic profile generation:** each concise profile uses the plain skill name (for example, "requirements gathering"). In the verbose profile, that is swapped for a longer description of the same skill (for example, "eliciting and formally capturing business needs from multiple departments"), and one general sentence about being a dedicated, detail-oriented professional is added. The phrase lists are in Annexure B.
- **User interface:** we did not build a web application. The system runs as a notebook and prints its results as tables, text and charts, which appear in Section 10, Section 11 and the Annexure.

10. Results and Evaluation

10.1 Matching quality on real data

Table 4 shows the average ranking metrics over the 3 Business Analyst job postings that had at least one Good Fit candidate.

Table 4: *Ranking results on real Business Analyst postings (K = 5, average of 3 jobs)*

Model	Precision@5	Recall@5	NDCG@5
A: TF-IDF + cosine	0.467	0.406	0.677
B: Embeddings + cosine	0.467	0.406	0.700

Both models found the same share of Good Fit candidates in the top 5. The embedding model had a slightly better NDCG@5, meaning it put better candidates a little higher up the list. The gap is small and comes from only 3 job postings (each with 10 or more candidates), so we are not claiming that embeddings are better. The results also show that neither model is close to perfect: on average, fewer than half of the top 5 candidates were Good Fit.

10.2 Writing-style audit on synthetic data

Table 5 summarises how the scores and ranks changed when the same 15 profiles were rewritten from concise to verbose (n = 15 pairs).

Table 5: *Writing-style audit results (n = 15 profile pairs)*

Measure	TF-IDF	Embeddings
Mean score, concise version	0.214	0.422
Mean score, verbose version	0.028	0.341
Mean absolute score change	0.187	0.080
Profiles whose rank changed	14 of 15 (93.3%)	12 of 15 (80.0%)
True skills found by the skill extractor in the verbose text	None (0 in all 15 profiles)	Same extractor

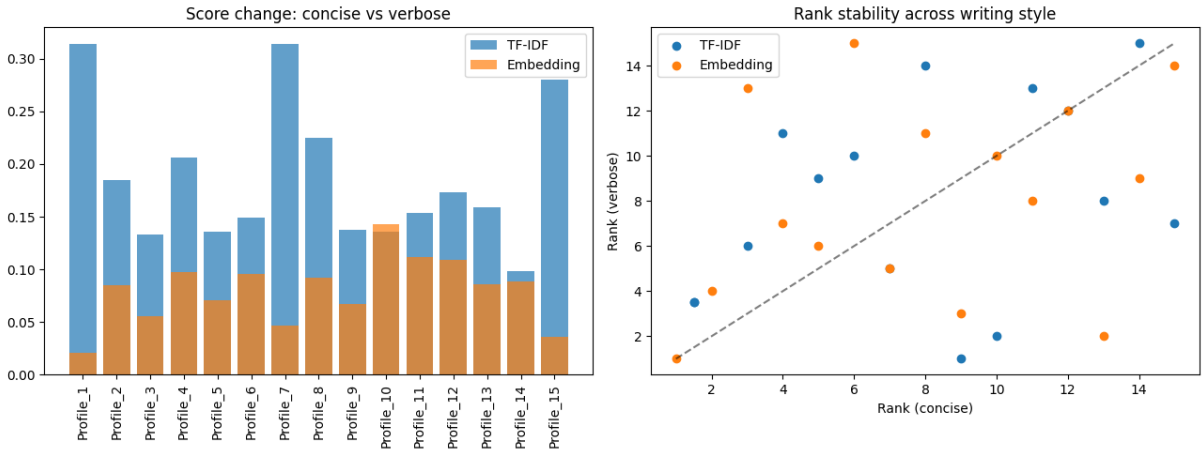


Figure 2: *Left: absolute score change between concise and verbose versions for each profile. Right: rank of each profile in the concise version (x-axis) against its rank in the verbose version (y-axis); points on the dashed line did not change rank.*

What the results mean.

- **TF-IDF is very sensitive to wording.** The average TF-IDF score fell from 0.214 to 0.028, a drop of about 87%, even though the skills did not change. The verbose text avoids the exact words used in the job description, so there is hardly any word overlap left.
- **Embeddings are more stable, but not stable enough.** Their average score fell from 0.422 to 0.341 (about 19%). The change for individual profiles ranged from 0.021 (Profile_1) to 0.143 (Profile_10).
- **Rankings changed for most profiles.** 14 of 15 profiles changed rank with TF-IDF and 12 of 15 with embeddings. Only Profile_6 kept the same TF-IDF rank.
- **Stable scores and correct skill matching are different things.** The embedding model changed less, but the skill extractor, which is also part of the system, failed completely on the verbose text. In every one of the 15 profiles the number of missed skills equalled the number of true skills (for example, 8 missed out of 8 for Profile_1).

Answer to the research question. In this controlled experiment, yes. The way the skills were written changed the match score and the ranking even though the qualifications were equal, and the effect was bigger for TF-IDF than for the embedding model.

11. Error Analysis / NLP Failure Analysis

Case 1: Same skill, different words, skill not found. The dictionary looks only for fixed phrases, so a correct description in other words is missed. Table 6 shows examples from the synthetic data.

Table 6: *Examples where the verbose wording hides a true skill*

True skill	Concise wording	Verbose wording (skill missed)
requirements gathering	requirements gathering	eliciting and formally capturing business needs from multiple departments

True skill	Concise wording	Verbose wording (skill missed)
agile	agile	working within iterative sprint-based delivery frameworks
excel	excel	advanced spreadsheet modeling and reporting tools
data analysis	data analysis	interpreting large datasets to uncover actionable business insights

A human reader sees the same skill in each pair, but a phrase match does not. Profile_1 (all 8 skills), for example, had a TF-IDF score of 0.349 in the concise version and 0.035 in the verbose version. That is the largest TF-IDF drop (0.314), shared with Profile_7, which has the same skills. Its rank moved from 1.5 (tied) to 3.5 (tied).

Case 2: Embedding scores also fall. For Profile_10 the embedding score dropped from 0.459 to 0.316 (a change of 0.143), the largest embedding change, and its rank fell from 3 to 13. A likely reason is that the verbose version is longer and adds general phrases like "dedicated professional" and "proactive, detail-oriented" that are not in the job description, which pulls its vector away from the job vector. This is only our interpretation, since we did not test it separately.

Case 3: Equal skills, very different rank from a tiny change. Profile_1 and Profile_7 have the same skills and the same template, and as far as we can tell they differ only in the profile label in the text. Even so, their concise embedding scores were 0.404 and 0.465, which gave them ranks of 13 and 1. So a single token can move a candidate a long way in the ranking. It also shows that embedding scores are not fully stable either, and that profile labels or names should be removed from the text in future tests.

Case 4: High skill coverage but not a fit (real data). In the ServiceNow Business Analyst posting, the top 5 candidates by TF-IDF score included only 1 Good Fit candidate and 4 No Fit candidates. One of the No Fit candidates matched almost every required skill in our dictionary (for example process improvement, stakeholder management, communication, documentation, agile and Excel), while the Good Fit candidate matched six. A high skill count therefore does not mean a good match. The dictionary ignores years of experience (the job asks for 8+ years), tools such as ServiceNow, and industry.

Case 5: Possible false skill matches. The skill extractor uses substring matching, so the alias "excel" would also match the word "excellent", and "sql" would match inside other words. That can count a skill that is not really there. We found this by reading through the rules and did not measure how often it happened.

12. Ethical Considerations and Limitations

- **Fairness:** writing style can depend on a person's education, language background and culture, so a system that rewards the wording of the job description can disadvantage some groups. Our audit tests only one source of unfairness (wording). It does not look at gender, caste, age, region or any other personal attribute, and SkillSignal does not remove hiring bias.

- **Not a hiring decision:** the scores measure text similarity only. They are not a probability of being hired, and they say nothing about candidate quality or job performance. They should support a human recruiter, not replace one.
- **Privacy:** resumes contain personal information. We used a public dataset, but any real deployment would need consent, secure storage and careful handling of personal data.
- **Small and synthetic audit:** we used only 15 fictional profiles and one job description. The verbose versions also add a general filler sentence, so part of the score drop may come from length and not only from the changed skill wording. Some profiles have identical skills and tied scores, which can affect the rank-change counts. The results should not be generalised to all real applicants.
- **Limited real-data evaluation:** we used only Business Analyst postings, and only 3 jobs could be evaluated. The labels come from the dataset and we did not check them ourselves. Table 4 is therefore only a rough indication.
- **Method limits:** the skills dictionary is small (10 skills for the real data, 8 for the audit) and relies on exact phrases. Text was cut to 3,000 characters. TF-IDF was fitted on small groups, so its weights depend on the group. And the embedding model cannot be fully explained.

13. Conclusion

We built SkillSignal, a notebook-based resume–job matching system. It has two models (TF-IDF and sentence embeddings), a skills dictionary that reports matched and missing skills, a ranking evaluation on real data, and a writing-style audit on paired synthetic profiles. On real Business Analyst postings the two models performed equally on Precision@5 and Recall@5 (0.467 and 0.406), with embeddings slightly ahead on NDCG@5 (0.700 against 0.677), although the sample was small. In the audit, the same skills written in a different style changed scores by 0.187 (TF-IDF) and 0.080 (embeddings) on average, and changed the rank of 93.3% and 80.0% of the profiles. The skill extractor missed all skills in the verbose text. We think the objectives of the project were met at a small scale. What we learned is that text-similarity scores depend heavily on wording, that stable scores do not guarantee correct skill matching, and that tools like these should be audited before anyone trusts them.

14. Future Scope

- Build a larger skill vocabulary with synonyms and word-boundary matching, or use a skill-extraction model, so that skills described in other words get found too.
- Create more profiles and several job families, and write the rewrites by hand or in several different styles, so that the effects of length and style can be separated.
- Strip names and labels from the text and test them as a separate factor.
- Evaluate on more job postings and add more metrics, such as confidence intervals across jobs.
- Try other embedding models, and fine-tune one so that paraphrases of the same skill get similar scores.

- Build a simple recruiter interface that shows scores, matched and missing skills, and a warning that the score is not a hiring decision.

15. References

[1] cnamuangtoun, "resume-job-description-fit," Hugging Face Datasets. [Online]. Available: https://huggingface.co/datasets/cnamuangtoun/resume-job-description-fit

[2] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. EMNLP-IJCNLP, 2019.

[3] W. Wang et al., "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020.

[4] Sentence-Transformers, "all-MiniLM-L6-v2" model card, Hugging Face. [Online]. Available: https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

[5] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," Information Processing & Management, vol. 24, no. 5, pp. 513–523, 1988.

[6] K. Järvelin and J. Kekäläinen, "Cumulated gain-based evaluation of IR techniques," ACM Trans. Information Systems, vol. 20, no. 4, pp. 422–446, 2002.

[7] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[8] Q. Lhoest et al., "Datasets: A community library for natural language processing," in Proc. EMNLP (System Demonstrations), 2021.

[9] W. McKinney, "Data structures for statistical computing in Python," in Proc. 9th Python in Science Conf. (SciPy), 2010.

[10] J. D. Hunter, "Matplotlib: A 2D graphics environment," Computing in Science & Engineering, vol. 9, no. 3, pp. 90–95, 2007.

16. GitHub Repository

The notebook with the complete code and this report can be found in the repository below.

- **GitHub Repository Link:** https://github.com/ayush207/NLP-Project-Ayush-Yana-Anushka-

17. Team Contribution

The table below lists what each member worked on.

Team Member	PRN	Specific Contribution
Ayush Sahoo	25030422031	Model
Anushka Thakur	25030422019	PPT
Yana Khandelwal	25030422141	Model Card

18. Annexure

Annexure A: Skills dictionary used for real data (Dataset A)

Table A1: *Skills and the phrases used to detect them*

Skill	Phrases searched
sql	sql
excel	excel
requirements gathering	requirements gathering, requirement gathering, business requirement
stakeholder management	stakeholder, stakeholders
agile	agile, scrum
documentation	documentation, use cases, user stories
process improvement	process improvement, process mapping, process optimization
communication	communication, communicate
project management	project management, project plan
data analysis	data analysis, analytical, analytics

For the audit (Dataset B), the 8 skills used were the concise phrases themselves: requirements gathering, stakeholder management, process improvement, agile, excel, documentation, communication and data analysis.

Annexure B: Phrase pairs used to create the synthetic profiles

Table A2: *Concise and verbose wording for each skill*

Skill	Concise phrase	Verbose phrase
requirements gathering	requirements gathering	eliciting and formally capturing business needs from multiple departments
stakeholder management	stakeholder management	building strong working relationships with cross-functional partners and leadership
process improvement	process improvement	redesigning existing workflows to increase operational efficiency
agile	agile	working within iterative sprint-based delivery frameworks
excel	excel	advanced spreadsheet modeling and reporting tools

Skill	Concise phrase	Verbose phrase
documentation	documentation	producing detailed functional specifications and user stories
communication	communication	clearly conveying complex findings to both technical and non-technical audiences
data analysis	data analysis	interpreting large datasets to uncover actionable business insights

Job description used for Dataset B: "We are looking for a Business Analyst skilled in requirements gathering, stakeholder management, and process improvement. The candidate should be experienced with Agile methodology, strong in Excel, capable of producing clear documentation, and possess excellent communication skills. Data analysis experience is a plus."

Annexure C: Score and rank results for all 15 profile pairs

Table A3: *Scores for each profile (change = absolute difference; n = 15)*

Profile	No. of skills	TF-IDF concise	TF-IDF verbose	TF-IDF change	Embed concise	Embed verbose	Embed change	Rank changed (TF-IDF / Emb.)
Profile_1	8	0.349	0.035	0.314	0.404	0.383	0.021	Yes / Yes
Profile_2	4	0.211	0.026	0.185	0.409	0.324	0.085	Yes / Yes
Profile_3	4	0.162	0.029	0.133	0.417	0.362	0.055	Yes / Yes
Profile_4	3	0.234	0.027	0.206	0.434	0.337	0.097	Yes / Yes
Profile_5	3	0.172	0.037	0.135	0.380	0.309	0.071	Yes / Yes
Profile_6	3	0.170	0.021	0.149	0.462	0.366	0.096	No / Yes
Profile_7	8	0.349	0.035	0.314	0.465	0.418	0.047	Yes / No
Profile_8	4	0.250	0.025	0.225	0.414	0.322	0.092	Yes / Yes
Profile_9	3	0.174	0.037	0.137	0.391	0.323	0.067	Yes / Yes
Profile_10	4	0.152	0.017	0.136	0.459	0.316	0.143	Yes / Yes
Profile_11	3	0.171	0.018	0.154	0.420	0.308	0.112	Yes / Yes
Profile_12	5	0.206	0.032	0.173	0.439	0.330	0.109	Yes / Yes
Profile_13	3	0.176	0.017	0.159	0.409	0.323	0.086	Yes / No
Profile_14	3	0.129	0.031	0.098	0.408	0.319	0.089	Yes / No
Profile_15	8	0.312	0.032	0.280	0.412	0.376	0.036	Yes / Yes

Annexure D: Sample candidate match output (real data)

Output printed for the ServiceNow Business Analyst posting (largest evaluated group). Scores are cosine similarities, not hiring probabilities. Only the top 3 of the 5 printed candidates are shown, and the skill lists were partly cut off in the notebook printout.

Table A4: *Top candidates by TF-IDF score for one job (real data)*

Dataset label	TF-IDF	Embedding	Matched skills shown in the output

Dataset label	TF-IDF	Embedding	Matched skills shown in the output
Good Fit	0.213	0.588	stakeholder management, communication, documentation, agile, excel, requirements gathering
No Fit	0.190	0.521	communication, excel, process improvement, data analysis
No Fit	0.149	0.503	process improvement, stakeholder management, communication, documentation, agile, excel, requirements gathering (list cut off)